

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	11
Experiment Name	Time Series Analysis of Website Traffic

DESCRIPTION

To perform the experiment: Time Series Analysis of Website Traffic.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Date": pd.date_range("2025-01-01", periods=10, freq="D"),
    "Page_Views": [100, 110, 105, 120, 125, 130, 128, 135, 300, 140]
}).set_index("Date")
df["Moving_Average"] = df["Page_Views"].rolling(3, min_periods=1).mean()
print(df)
df[["Page_Views", "Moving_Average"]].plot(marker="o")
plt.title("Website Traffic and 3-Day Moving Average")
plt.ylabel("Page Views"); plt.tight_layout(); plt.show()
```

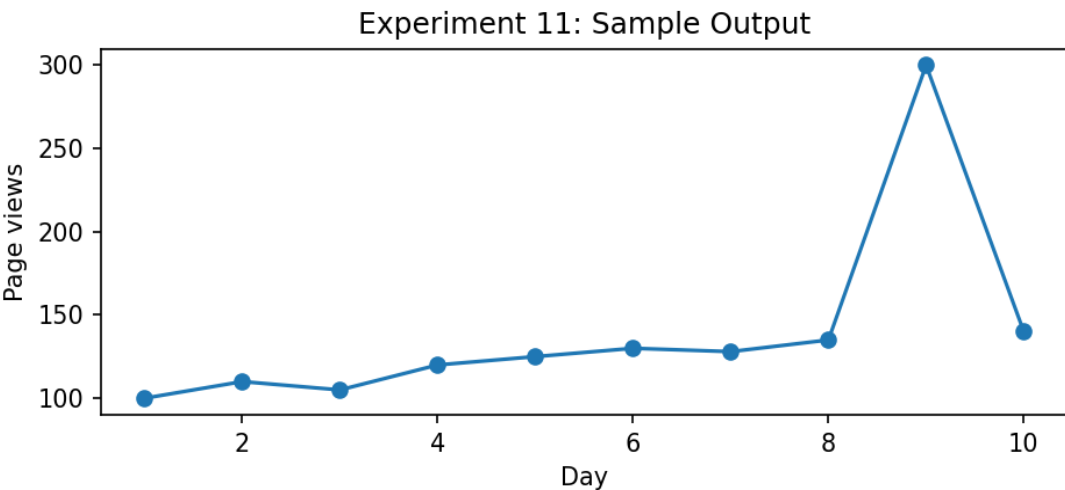
SAMPLE INPUT

Sample input: ten daily page-view counts, including a spike to 300.

OUTPUT — SAMPLE CONSOLE OUTPUT

Page_Views Moving_Average\n100 100.0\n110 105.0\n105 105.0\n... ..

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature
